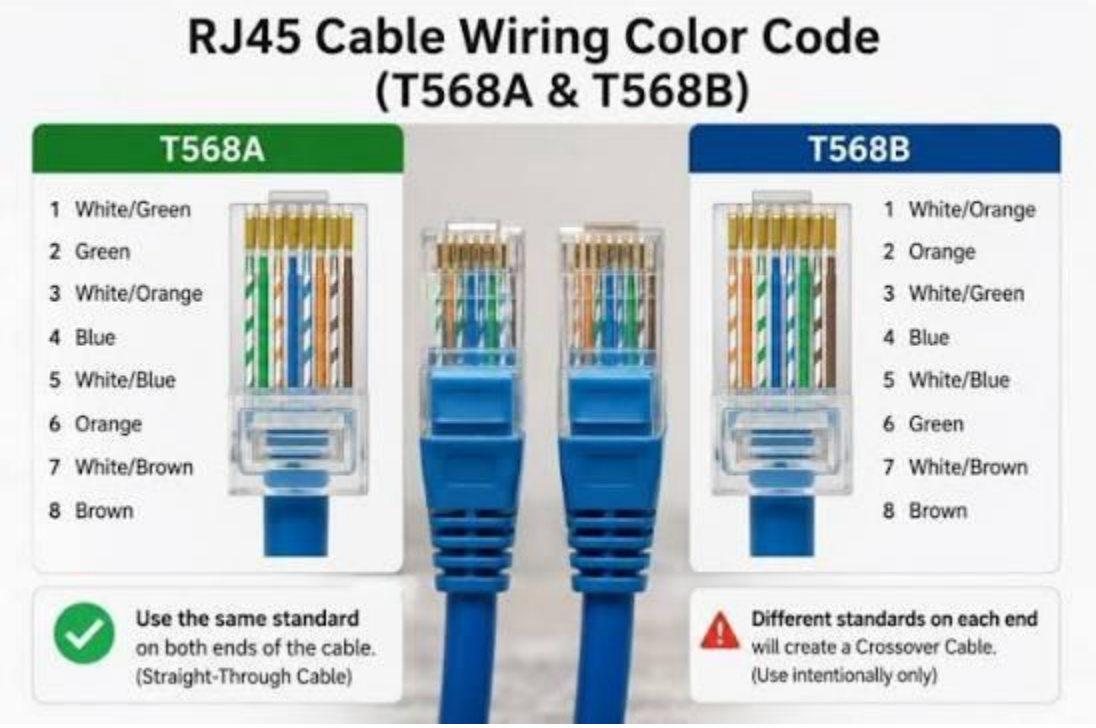
ETHERNET STANDARD AND SPECIFICATION

1. Draw a simple diagram showing the pin layout of an RJ-45 connector and label the color codes for the T568A and T568B Ethernet wiring standards.

Ans:-



- **T568A (Pin 1 to 8):** White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, and Brown.
- **T568B (Pin 1 to 8):** White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, and Brown.

2. Create a table comparing the key specifications of Ethernet standards 10BASE-T, 100BASE-TX, and 1000BASE-T, including speed, cable type, maximum cable length, and connector type.

Ans:-

Specification	10BASE-T	100BASE-TX	1000BASE-T
Common name	Ethernet	Fast Ethernet	Gigabit Ethernet
Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Cable type	Twisted pair (UTP), Cat3 or better	Twisted pair (UTP), Cat5 or better	Twisted pair (UTP), Cat5e or Cat6
Wire pairs used	2 pairs	2 pairs	4 pairs
Maximum cable length	100 metres	100 metres	100 metres
Connector type	RJ-45	RJ-45	RJ-45

3. Given a scenario where you need to connect two laptops directly using Ethernet, explain which cable (straight-through or crossover) you would use and why.

Hint: Think about how transmit and receive pairs are wired in each cable type.

   Ans:-

Cable chosen: Crossover cable

Why: Both laptops send data through pins 1 and 2 and receive data through pins 3 and 6. A straight-through cable connects the same pins, so both laptops try to send data on the same pins. A crossover cable swaps the transmit and receive connections, allowing both laptops to communicate.

Pin on Laptop A	Cable wire	Pin on Laptop B
Pin 1 (TX+)	goes to	Pin 3 (RX+)
Pin 2 (TX-)	goes to	Pin 6 (RX-)
Pin 3 (RX+)	goes to	Pin 1 (TX+)
Pin 6 (RX-)	goes to	Pin 2 (TX-)

Note: Most modern laptops have Auto MDI-X, which automatically adjusts the transmit and receive pairs, so even a straight-through cable usually works today. The crossover cable is the correct choice for older devices.

4. List three real-world devices in your home or college that use Ethernet connections, and for each, specify the likely Ethernet standard (e.g., 100BASE-TX, 1000BASE-T) they support.

Ans:-

Device	Where	Likely Ethernet standard	Why
WiFi router	Home / college	1000BASE-T	Modern routers have gigabit LAN ports to match fast internet plans.
Desktop PC	College computer lab	1000BASE-T	Most current PCs and laptops have a gigabit Ethernet port.
Smart TV	Home	100BASE-TX	Many smart TVs have a 100 Mbps port, which is enough for HD/4K streaming.